

Compact-Image Obstructions for a Hyperbolic Grading in $\mathrm{Sp}(32, 32)$

Neutral, grading-even, and extremal order parameters

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Abstract

Let a connected real reductive group G act faithfully on a finite-dimensional real vector space E , and let $A_Z = \exp(\mathbb{R}Z)$ be a one-parameter subgroup. We study whether the stabilizer of a specified order parameter has relatively compact image in $\mathrm{GL}(E)$, equivalently whether it preserves a positive-definite inner product on E . We prove three elementary compact-image obstructions. An A_Z -invariant vector retains the entire one-parameter group in its stabilizer. An operator commuting with $z = d\iota(Z)$ retains the same group in its centralizer. For the extremal leg, assume that $\mathrm{ad}(Z)$ and $dR(Z)$ are diagonalizable over $\mathbb{R}$. If the positive, respectively negative, $\mathrm{ad}(Z)$ -eigenspace sum contains X_+ , respectively X_- , whose image under $d\iota$ is nonzero nilpotent, then a maximal, respectively minimal, Z -weight vector retains the corresponding unipotent one-parameter subgroup. Whenever the retained subgroup has unbounded represented image, no invariant positive-definite inner product exists.

For $G = \mathrm{Sp}(32, 32)$ we verify the hypotheses directly in the defining quaternionic matrix model. The relevant grading has only $\mathrm{ad}(Z)$ -eigenvalues $-2, 0, 2$; the two nonzero eigenspaces have real dimension 2080, are abelian, and consist of square-zero matrices. When $z^2 = 1$, conjugation by z separately defines a discrete involution on $\mathrm{End}_{\mathbb{R}}(E)$, and a grading-even operator is precisely a Z -commuting operator. We do not identify continuous A_Z -invariance of an arbitrary vector representation with preservation of a discrete grading.

The ingredients are standard compact-group averaging, linear algebra, and weight raising. The contribution is a self-contained synthesis for one fixed hyperbolic grading, an explicit quaternionic specialization, and a diagnostic separation of three commonly conflated notions: continuous neutrality, discrete operator parity, and extremal weight. The result is finite-dimensional and algebraic; it does not establish an interacting Hilbert space, a vacuum, a dynamical condensate, or physical unitarity.

1. Result at a glance

The mathematical obstruction has three legs with deliberately separate hypotheses:

order parameter	subgroup forced into its stabilizer	compact-image obstruction
vector fixed by A_Z	A_Z	the tested image of A_Z is unbounded
operator O with $[O, d\iota(Z)] = 0$	$A_Z \subset C_G(O)$	the same unbounded image
maximal Z -weight vector, with real-diagonalizable $\mathrm{ad}(Z)$ and $dR(Z)$ and nonzero nilpotent $d\iota(X_+)$	the corresponding positive unipotent one-parameter subgroup	nonconstant polynomial growth in the tested image
minimal Z -weight vector, with real-diagonalizable $\mathrm{ad}(Z)$ and $dR(Z)$ and nonzero nilpotent $d\iota(X_-)$	the corresponding negative unipotent one-parameter subgroup	nonconstant polynomial growth in the tested image

For the displayed $\mathrm{Sp}(32, 32)$ grading, all three mechanisms are visible in exact blocks. The standard compact-reducing involution has centralizer $\mathrm{Sp}(32) \times \mathrm{Sp}(32)$, but it does not commute with the grading generator.

The strongest conclusion is:

On the faithful finite-dimensional module tested here, vectors invariant under the unbounded grading flow, operators commuting with its infinitesimal generator, and extremal-weight vectors for real-diagonalizable $\mathrm{ad}(Z)$ and $dR(Z)$ with the corresponding sign-specific witness having nonzero nilpotent $d\iota$ -image cannot have relatively compact stabilizer image. When the generator is involutory on the tested module, the operator statement applies in particular to grading-even operators.

Four distinctions are load-bearing.

1. Compactness is asserted for the closure of the **represented image**, not for an abstract source group hidden behind a nonfaithful action.
2. A_Z -invariance is invariance under a continuous one-parameter subgroup. It is not automatically a discrete parity statement.
3. Grading-even is defined for operators only when $z = d\iota(Z)$ supplies an involution by conjugation.
4. Preservation of a positive-definite form is a finite-dimensional algebraic proxy. It is not a theorem about interacting quantum stability.

2. Mathematical setting

2.1 Representations and stabilizers

Let E be a finite-dimensional real vector space and let

$$\iota : G \hookrightarrow \mathrm{GL}(E)$$

be a faithful finite-dimensional representation of a connected real reductive group. The underlying real representation is used even when E originates as a complex or quaternionic module.

All group representations in this paper are finite-dimensional continuous, hence smooth, real Lie-group representations. Closures, boundedness, and relative compactness are taken in the usual Euclidean topology of the relevant real matrix space.

For a vector w in a finite-dimensional representation $R : G \rightarrow \mathrm{GL}(V)$, write

$$G_w = \{g \in G : R(g)w = w\}.$$

For a finite family $w_i \in V_i$ in representations R_i , its common vector stabilizer is

$$G_{\mathbf{w}} = \{g \in G : R_i(g)w_i = w_i \text{ for every } i\}.$$

For $O \in \mathrm{End}_{\mathbb{R}}(E)$ under conjugation, the stabilizer is the centralizer

$$C_G(O) = \{g \in G : \iota(g)O = O\iota(g)\}.$$

All compactness statements below concern the closure of the image under ι . Faithfulness is stronger than strictly necessary, but it prevents a noncompact stabilizer direction from disappearing in the module on which positivity is tested. The proofs require only that the named obstruction subgroups have unbounded image under ι .

2.2 Finite-dimensional invariant majorants

An **invariant positive majorant** for a subgroup $H \leq G$ means, in this paper, a positive-definite real inner product h on E satisfying

$$h(\iota(a)x, \iota(a)y) = h(x, y) \quad (a \in H).$$

In finite dimensions every positive-definite norm induces the same vector-space topology, and every bilinear form b is continuous for that topology. After rescaling the positive form, one may arrange the conventional domination inequality

$$|b(x, y)|^2 \leq h(x, x)h(y, y).$$

In this paper “majorant” means only the invariant positive-definite form defined above; no infinite-dimensional notion of maximal, admissible, or topology-defining majorant is intended. Those distinctions are substantive in general Krein-space theory [Bognár 1974, chs. III–VI].

Suppose additionally that E has a nondegenerate symmetric form η preserved by G . If h and η are both H -invariant, define the invertible h -self-adjoint operator A by

$$\eta(x, y) = h(x, Ay).$$

To see equivariance explicitly, write $T = \iota(a)$ for $a \in H$. Simultaneous h - and η -invariance gives

$$h(x, T^{-1}ATy) = h(Tx, ATy) = \eta(Tx, Ty) = \eta(x, y) = h(x, Ay),$$

so $T^{-1}AT = A$. Finite-dimensional functional calculus therefore gives

$$|A| = (A^2)^{1/2}, \quad C = A|A|^{-1}, \quad C^2 = 1, \quad \eta(x, Cy) = h(x, |A|y).$$

The commuting h -self-adjoint operators A and C satisfy

$$\eta(Cx, y) = h(Cx, Ay) = h(x, ACy) = \eta(x, Cy),$$

so C is η -self-adjoint, while $AC = |A|$ gives $\eta(x, Cx) = h(x, |A|x) > 0$ for $x \neq 0$. Conversely, suppose C is an H -equivariant, η -self-adjoint involution satisfying $\eta(x, Cx) > 0$ for every $x \neq 0$. Then

$$h_C(x, y) = \eta(x, Cy)$$

is symmetric because η is symmetric and C is η -self-adjoint, positive definite by hypothesis, and H -invariant because C commutes with H and η is invariant. This states all the conditions used in the fundamental-symmetry formulation rather than treating positivity of $\eta(\cdot, C\cdot)$ alone as its definition.

2.3 Continuous flow, hyperbolicity, and discrete operator parity

Fix $Z \in \mathfrak{g}$ and set

$$A_Z = \exp(\mathbb{R}Z), \quad z = d\iota(Z).$$

The hypothesis needed for the first two obstruction legs is simply that $\iota(A_Z)$ is unbounded. A sufficient condition is that z have a nonzero real eigenvalue: on a corresponding eigenvector, $\exp(tz)$ grows like $e^{t\lambda}$ in one time direction.

The extremal-weight leg needs a different hypothesis. We call Z **ad-hyperbolic** when $\text{ad}(Z)$ is diagonalizable over $\mathbb{R}$. Then

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathbb{R}} \mathfrak{g}_\alpha, \quad \mathfrak{g}_\alpha = \{X \in \mathfrak{g} : [Z, X] = \alpha X\},$$

and we set

$$\mathfrak{n}_+ = \bigoplus_{\alpha > 0} \mathfrak{g}_\alpha, \quad \mathfrak{n}_- = \bigoplus_{\alpha < 0} \mathfrak{g}_\alpha.$$

These are $\text{ad}(Z)$ -eigenspace sums. When Z lies in a chosen maximal split abelian subspace, they are sums of restricted-root spaces having the same sign on Z [Helgason 2001, chs. III, VI; Knapp 2002, ch. VI]. A single nonzero real eigenvalue of z does **not** imply this real decomposition; the two hypotheses must not be conflated.

For any representation R , a vector is Z -**neutral** when $dR(Z)w = 0$. Equivalently,

$$R(\exp tZ)w = \exp(t dR(Z))w = w,$$

so it is fixed by the continuous group A_Z .

Only the extremal leg requires a weight decomposition. When $dR(Z)$ is diagonalizable over $\mathbb{R}$, write

$$V = \bigoplus_{\lambda \in \mathbb{R}} V_\lambda, \quad V_\lambda = \{v : dR(Z)v = \lambda v\}.$$

A nonzero $w \in V_\lambda$ has maximal, respectively minimal, Z -weight when no weight greater, respectively smaller, than λ occurs.

A separate discrete operator grading exists when $z^2 = 1$ on E . In that case

$$\sigma_Z(O) = zOz^{-1} = zOz$$

is an involution on $\text{End}_{\mathbb{R}}(E)$. An operator is grading-even when $\sigma_Z(O) = O$, equivalently $[O, z] = 0$, and grading-odd when $\sigma_Z(O) = -O$, equivalently $\{O, z\} = 0$. No corresponding order-two action on an arbitrary vector representation is assumed. In particular, continuous A_Z -neutrality is not called discrete grading invariance unless an additional representation dictionary proves that identification.

3. Compactness and two visible witnesses

Proposition 1 (invariant positive form versus compact closure). Let $H \leq \text{GL}(E)$. An H -invariant positive-definite inner product on E exists if and only if $\overline{H}$ is compact.

Proof. If h is invariant, H lies in the orthogonal group $O(E, h)$, which is compact in finite dimensions. Hence $\overline{H}$ is compact.

Conversely, if $\overline{H}$ is compact, choose any positive-definite inner product h_0 and average it over normalized Haar measure:

$$h(x, y) = \int_{\overline{H}} h_0(ax, ay) d\mu(a).$$

The average is positive definite and $\overline{H}$ -invariant. $\square$

This elementary equivalence fixes the exact stability predicate used in the paper. It requires neither closedness nor reductivity of the original subgroup H .

Lemma 2 (unbounded one-parameter witnesses). Suppose $H \leq G$.

1. If $A_Z \subset H$ and $\iota(A_Z)$ is unbounded, then $\overline{\iota(H)}$ is not compact.
2. If H contains $\exp(tX)$ for every $t \in \mathbb{R}$, where $d\iota(X)$ is nonzero and nilpotent, then $\overline{\iota(H)}$ is not compact.

Proof. The first conclusion is immediate. In the second case, $\exp(td\iota(X))$ is a nonconstant polynomial in t , hence unbounded. $\square$

4. Three compact-image obstructions

The four clauses below implement three mechanisms and intentionally do not share one overstrong hypothesis.

Theorem 3 (continuous-neutral, commuting-operator, and extremal obstructions). Let G , E , ι , and Z be as above.

1. Assume $\iota(A_Z)$ is unbounded. If every member of a finite family $w_i \in V_i$ is Z -neutral in its representation R_i , then $G_{\mathbf{w}}$ has non-relatively-compact image under ι .
2. Assume $\iota(A_Z)$ is unbounded. If $O \in \text{End}_{\mathbb{R}}(E)$ satisfies $[O, z] = 0$, then $C_G(O)$ has non-relatively-compact image under ι .
3. Assume Z is ad-hyperbolic, $dR(Z)$ is diagonalizable over $\mathbb{R}$, and $w \neq 0$ is a maximal Z -weight vector. If $\mathfrak{n}_+$ contains X_+ for which $d\iota(X_+)$ is nonzero nilpotent, then G_w has non-relatively-compact image under ι .
4. Assume Z is ad-hyperbolic, $dR(Z)$ is diagonalizable over $\mathbb{R}$, and $w \neq 0$ is a minimal Z -weight vector. If $\mathfrak{n}_-$ contains X_- for which $d\iota(X_-)$ is nonzero nilpotent, then G_w has non-relatively-compact image under ι .

Consequently none of these stabilizers preserves a positive-definite inner product on E .

Proof.

1. Since $dR_i(Z)w_i = 0$,

$$R_i(\exp tZ)w_i = \exp(tdR_i(Z))w_i = w_i$$

for every i and t . Hence $A_Z \subset G_{\mathbf{w}}$. Lemma 2 and Proposition 1 apply.

2. The commutator condition gives

$$O \exp(tz) = \exp(tz)O.$$

Since $\iota(\exp tZ) = \exp(tz)$, we have $A_Z \subset C_G(O)$. Apply Lemma 2 and Proposition 1.

3. Suppose $dR(Z)w = \lambda w$ and λ is maximal. For $X \in \mathfrak{g}_{\alpha}$ with $\alpha > 0$,

$$dR(Z)dR(X)w = (\lambda + \alpha)dR(X)w.$$

No weight above λ occurs, so $dR(X)w = 0$. By linearity, $dR(\mathfrak{n}_+)w = 0$, in particular $dR(X_+)w = 0$. Therefore

$$R(\exp tX_+)w = \exp(tdR(X_+))w = w,$$

and $\exp(\mathbb{R}X_+) \subset G_w$. Its image under ι is unbounded by Lemma 2. Proposition 1 completes the maximal case.

4. For a minimal weight the same calculation with $\alpha < 0$ shows that $dR(\mathfrak{n}_-)w = 0$. In particular X_- exponentiates to a subgroup of G_w whose represented image is unbounded by Lemma 2. Proposition 1 completes the minimal case. $\square$

Remark 4 (vector versus line stabilizers). A parabolic subgroup generally stabilizes the extremal line $\mathbb{R}w$ rather than the chosen vector. The proof uses only a positive or negative nilpotent subgroup. Its Lie algebra annihilates w , so exponentiation fixes w pointwise. The vector-stabilizer claim is therefore stronger than, and not confused with, line stabilization.

Remark 5 (minimal hypotheses). Faithfulness can be weakened if the tested representation still detects the named unbounded subgroup. Likewise, parts 3–4 each require only one represented nonzero nilpotent witness of the appropriate sign, not algebraicity of every representation or a classification of every root space.

5. Compact-reducing operators and parity

Let $P' \in \text{End}_{\mathbb{R}}(E)$ be an involution such that $C_G(P')$ has relatively compact represented image. This is the only property needed for the next result.

Corollary 6 (a compact-reducing involution cannot commute with the grading flow). If $\iota(A_Z)$ is unbounded, then $[P', z] \neq 0$.

Proof. If $[P', z] = 0$, then $A_Z \subset C_G(P')$, contradicting relative compactness by Lemma 2. $\square$

In a classical matrix realization, an additional hypothesis is needed before calling P' a Cartan implementer: conjugation by P' must normalize G (or its Lie algebra) and implement a Cartan involution there. The compactness argument itself does not infer this structure from involutivity alone.

Corollary 6 proves non-invariance, not pure oddness. If $z^2 = 1$, every operator has the decomposition

$$P' = P'_{\text{even}} + P'_{\text{odd}}, \quad P'_{\text{even}} = \frac{1}{2}(P' + zP'z), \quad P'_{\text{odd}} = \frac{1}{2}(P' - zP'z).$$

The condition $[P', z] \neq 0$ says only that the odd component is nonzero. It does not say that the even component vanishes. Pure oddness follows only from the additional homogeneous condition $zP'z = -P'$.

6. The $\text{Sp}(32, 32)$ specialization

6.1 Quaternionic matrix model

Let $n \geq 1$ and regard $E = \mathbb{H}^n \oplus \mathbb{H}^n$ as a right quaternionic vector space, with quaternionic matrices acting on the left and $*$ denoting conjugate transpose. Unless otherwise stated, $\text{End}(E)$ means $\text{End}_{\mathbb{R}}(E)$ on the $8n$ -dimensional underlying real space. The calculation below is valid for every $n \geq 1$; the numerical specialization of interest is $n = 32$, where $\dim_{\mathbb{R}} E = 256$. Set

$$\beta = \begin{pmatrix} I_n & 0 \\ 0 & -I_n \end{pmatrix}.$$

The real Lie group

$$G = \text{Sp}(n, n) = \{g \in \text{GL}_{2n}(\mathbb{H}) : g^* \beta g = \beta\}$$

is a connected real reductive Lie group and acts faithfully on the underlying real vector space of E [Sedano-Mendoza 2019]. Its Lie algebra is

$$\mathfrak{g} = \left\{ X = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix} : A^* = -A, D^* = -D \right\}.$$

We use the standard notation and real-form convention of [Sedano-Mendoza 2019]. The maximal compact subgroup is

$$K = \text{Sp}(n) \times \text{Sp}(n).$$

Since $\dim_{\mathbb{R}} \text{Sp}(m) = m(2m + 1)$,

$$\dim G = (2n)(4n + 1),$$

$$\dim K = 2n(2n + 1), \quad \dim(G/K) = 4n^2.$$

At $n = 32$ these are 8256, 4160, and 4096, respectively.

The group preserves the nondegenerate real symmetric form

$$\eta(x, y) = \text{Re}(x^* \beta y).$$

For the compact reducer $P = \beta$ used below, $h_P(x, y) = \eta(x, Py) = \text{Re}(x^*y)$ is positive definite. Thus the abstract fundamental-symmetry construction of Section 2.2 and the concrete quaternionic model use the same positivity predicate.

6.2 The grading generator and a compact reducer

Define

$$z = d\iota(Z) = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}.$$

Direct multiplication gives

$$z^*\beta + \beta z = 0,$$

so $z \in \mathfrak{sp}(n, n)$, and $z^2 = I$. Moreover,

$$\exp(tz) = \begin{pmatrix} \cosh(t)I_n & \sinh(t)I_n \\ \sinh(t)I_n & \cosh(t)I_n \end{pmatrix}$$

has eigenvalues e^t and e^{-t} . Thus $\iota(A_Z)$ is unbounded. The continuous group A_Z and the discrete conjugation involution $O \mapsto zOz$ are related through the same infinitesimal operator but are not the same action.

The standard compact-reducing involution is $P = \beta$. It satisfies

$$P^2 = I, \quad Pz = -zP.$$

An element commuting with P is block diagonal, and intersecting the block-diagonal group with $g^*\beta g = \beta$ gives

$$C_G(P) = \text{Sp}(n) \times \text{Sp}(n).$$

Here conjugation by P normalizes G and implements the standard Cartan involution. This particular P is genuinely grading-odd. Corollary 6 gives the reducer-independent statement relative to this fixed z : any involution with relatively compact centralizer fails to be grading-even, although a conjugate representative need not be purely odd in the fixed frame.

6.3 Exact $\text{ad}(Z)$ decomposition

The isotropic basis makes the structure transparent. Let

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} I_n & I_n \\ I_n & -I_n \end{pmatrix}, \quad J = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}.$$

Then

$$S^{-1}zS = \begin{pmatrix} I_n & 0 \\ 0 & -I_n \end{pmatrix}, \quad S^*\beta S = J.$$

Writing $Y = S^{-1}XS = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the Lie condition $Y^*J + JY = 0$ is equivalent to

$$d = -a^*, \quad b^* = -b, \quad c^* = -c.$$

Consequently

$$\mathfrak{g}_2 = \left\{ \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} : b^* = -b \right\}, \quad \mathfrak{g}_0 = \left\{ \begin{pmatrix} a & 0 \\ 0 & -a^* \end{pmatrix} : a \in M_n(\mathbb{H}) \right\} \cong \mathfrak{gl}(n, \mathbb{H}),$$

$$\mathfrak{g}_{-2} = \left\{ \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} : c^* = -c \right\}.$$

This immediately exhibits the nonzero eigenspaces as abelian and square-zero. It also identifies $\mathfrak{p}_Z = \mathfrak{g}_0 \oplus \mathfrak{g}_2$ as the parabolic stabilizing the first, isotropic quaternionic n -plane, with abelian nilradical $\mathfrak{g}_2$: in the S -basis, its matrices are exactly the displayed upper-triangular blocks and hence preserve $\mathbb{H}^n \oplus 0$, while the lower-left block detects failure to preserve that plane. Rescaling the grading element by $1/2$ turns the displayed degrees into $-1, 0, 1$, so this is a $|1|$ -grading rather than a depth-two $|2|$ -grading [Čap and Slovák 2009, chs. 2–3].

Conjugating this decomposition back to the original basis gives the following blocks and fixes every sign explicitly.

For

$$X = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix} \in \mathfrak{sp}(n, n),$$

one computes

$$[z, X] = \begin{pmatrix} B^* - B & D - A \\ A - D & B - B^* \end{pmatrix}.$$

Solving $[z, X] = 2X$ gives

$$X_+(B) = \begin{pmatrix} -B & B \\ -B & B \end{pmatrix}, \quad B^* = -B.$$

Solving $[z, X] = -2X$ gives

$$X_-(B) = \begin{pmatrix} B & B \\ -B & -B \end{pmatrix}, \quad B^* = -B.$$

Finally $[z, X] = 0$ gives

$$X_0(A, B) = \begin{pmatrix} A & B \\ B & A \end{pmatrix}, \quad A^* = -A, \quad B^* = B.$$

Therefore

$$\mathfrak{sp}(n, n) = \mathfrak{g}_{-2} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_2.$$

A skew-Hermitian quaternionic $n \times n$ matrix has real dimension

$$3n + 4 \binom{n}{2} = n(2n + 1),$$

while a Hermitian one has real dimension

$$n + 4 \binom{n}{2} = n(2n - 1).$$

At $n = 32$ this yields

$$\dim \mathfrak{g}_2 = \dim \mathfrak{g}_{-2} = 32(65) = 2080,$$

$$\dim \mathfrak{g}_0 = 32(65) + 32(63) = 4096,$$

and $2080 + 4096 + 2080 = 8256$ as required.

The decisive nilpotence is equally explicit. For all skew-Hermitian B, C ,

$$X_+(B)X_+(C) = 0, \quad X_-(B)X_-(C) = 0.$$

Thus $\mathfrak{g}_2$ and $\mathfrak{g}_{-2}$ are nonzero abelian subalgebras whose elements are square-zero in the defining representation. In particular,

$$\exp(tX_{\pm}(B)) = I + tX_{\pm}(B)$$

is unbounded for $B \neq 0$. These eigenspaces are the positive and negative $\text{ad}(Z)$ -eigenspace sums; relative to a maximal split abelian subspace containing Z , they are sums of restricted-root spaces. This block calculation supplies every sign-specific witness hypothesis used by Theorem 3(3)–(4) without appealing to a toy model or to an unspecified algebraicity assumption.

Corollary 7 (the $\text{Sp}(n, n)$ family and the $\text{Sp}(32, 32)$ specialization). For every $n \geq 1$ and the z above, the common stabilizer of any finite family of A_Z -invariant vectors in finite-dimensional smooth representations has non-relatively-compact image under the defining representation. The same holds for the centralizer of every grading-even operator on E . If R is a finite-dimensional representation with $dR(Z)$ diagonalizable over $\mathbb{R}$, the conclusion also holds for every nonzero maximal or minimal Z -weight vector: the explicitly displayed X_+ or X_- supplies the required nonzero nilpotent witness of the corresponding sign. At $n = 32$, the positive and negative witness spaces each have real dimension 2080.

7. Why the scope conditions are necessary

7.1 One real eigenvalue is not full hyperbolicity

In a product of a split group and a compact nonabelian group, a semisimple element can have a split component and an elliptic component. A faithful direct sum representation then has a nonzero real eigenvalue and also nonreal eigenvalues in its complexified spectrum; on the underlying real space the elliptic part appears in rotation blocks and is not real diagonalizable. Its one-parameter image is unbounded, so Theorem 3(1) and (2) can still apply, but a real $\text{ad}(Z)$ -eigenspace decomposition for parts (3)–(4) does not follow. This is why the theorem states the two hypotheses separately.

Nor does ad -hyperbolicity create a nilpotent witness. For $G = \mathbb{R}_{>0}$ acting on $E = \mathbb{R}$ by scalar multiplication, $\text{ad}(Z) = 0$ is real diagonalizable but $\mathfrak{n}_+ = \mathfrak{n}_- = 0$; the nonzero vector 1 is both maximal and minimal and has trivial, hence compact, stabilizer. The sign-specific witness hypothesis is therefore indispensable.

7.2 Not every charged vector has noncompact stabilizer

Theorem 3 does not cover a non-extremal weight vector or a vector that is not a Z -eigenvector. This exclusion is substantive. A timelike vector in the standard representation of $\text{SO}(2, 1)$ has compact stabilizer $\text{SO}(2)$ but is not an eigenvector of a boost mixing its time and space coordinates. Thus the stronger slogan “every boost-charged vector has noncompact stabilizer” is false.

7.3 Compact subgroups need not use one preferred frame

Maximal compact subgroups of a connected real reductive group are conjugate, not identical. A compact subgroup can lie in a tilted conjugate gKg^{-1} without commuting with one preferred Cartan implementer. Neither Theorem 3 nor Corollary 6 uses the false implication

$$\text{compact image of } H \implies \iota(H) \subset C_{\text{GL}(E)}(P)$$

for a fixed P .

8. Scope and interpretation boundary

The theorem is entirely a finite-dimensional statement about represented stabilizers. “Order parameter” means a chosen vector, family of vectors, or operator on which a stabilizer is taken. It does not assert that the object is a dynamically formed condensate or a vacuum expectation value.

If a physical model identifies an involutory z with a chirality or mirror grading, Corollary 7 says that a grading-even operator cannot simultaneously produce a compact-image centralizer in this defining module. Applying the vector legs to the same physical grading requires an additional dictionary identifying the relevant vector representation and its A_Z action. The paper does not supply or assume such a physical dictionary.

The compact boundary is:

- interacting unitarity or ghost clearance;
- a physical state space, observable algebra, or scattering operator;
- existence or dynamical selection of a vacuum;
- anomaly cancellation, a mass spectrum, or chirality of a field theory;
- compactification of spacetime or extra dimensions;
- completeness of any model-specific order-parameter census;
- a theorem for non-extremal or mixed-weight vectors, nonfaithfully detected subgroups, or infinite-dimensional metric operators and domains.

These are not hidden premises of the proof. They are different problems.

9. Escape taxonomy and falsifiers

The theorem is diagnostic: a proposed escape should name the hypothesis it changes.

1. **Bounded flow.** Show that the tested image of A_Z is bounded. This defeats parts (1) and (2), but it is false for the displayed $\text{Sp}(32, 32)$ generator.
2. **Non-invariant vector.** Use a vector not fixed by A_Z and not extremal for the Z -weight decomposition.
3. **Noncommuting operator.** Use an operator that does not commute with z . A compact reducer must take this route by Corollary 6.
4. **No represented nilpotent witness.** Exhibit a setting in which the relevant positive or negative eigenspace is absent or invisible in the majorant-tested representation.
5. **Different positivity predicate.** Replace invariant positive-definite inner product on E with a precisely defined physical construction not equivalent to compact represented closure.
6. **Infinite-dimensional exit.** Supply domains, closures, topologies, and boundedness hypotheses adequate for an infinite-dimensional theorem.

A counterexample to any written implication would falsify the theorem. A model lying outside one of the quantified classes is a scope exit, not a counterexample.

10. Machine-checked hardening and reproducibility

The written proofs are load-bearing. The repository candidate includes three paper-specific machine-checking layers:

1. An exact SageMath certificate constructs rational quaternionic matrices and checks Lie-algebra membership, the ± 2 commutator equations, mutual square-zero products, the exact isotropic basis change and triangular block forms, nonzero linear truncation witnesses, and all dimension formulas. Together with $\exp(tX) = I + tX$ from $X^2 = 0$, the nonzero witnesses certify the stated polynomial growth.
2. A property-based Python certificate generates exact small-rank quaternionic matrices, checks the same identities without floating point, and includes planted sign and parity mutations that the test suite must reject.
3. A Lean 4 module checks a narrow algebraic kernel: conjugation-fixed versus commuting parity for an involutory element, the shifted-eigenvector identity behind weight raising, extremal annihilation under an explicit no-higher-weight hypothesis, and the square-zero block maps.

The Lean module does **not** formalize Proposition 1, Haar averaging, the full theory of real reductive groups, quaternionic Lie-group membership, or the physical interpretation. The Sage and property tests are executable certificates, not substitutes for the general proof. Exact file paths, commands, dependency locks, theorem names, expected outputs, and axiom dependencies appear in `REPRODUCE.md` and `VERIFICATION.md` beside this manuscript. An archival submission must include the complete versioned evidence tree listed below and its checksum manifest. If this Markdown file is circulated alone, the machine checks are unavailable to that reader and should receive no evidentiary weight.

11. Prior art, novelty, and use

Every general ingredient is established mathematics:

- invariant inner products can be averaged over compact groups;
- Cartan involutions determine maximal compact subgroups, unique up to conjugacy;
- a nonzero real weight makes a one-parameter group unbounded in the tested representation;
- extremal-weight directions are annihilated by the appropriate raising or lowering algebra;
- a one-parameter subgroup with nonzero nilpotent image is unbounded in the tested representation.

Compact averaging and Cartan/restricted-root structure are standard in [Helgason 2001, chs. III, VI, IX; Knapp 2002, chs. IV, VI]. Indefinite-inner-product terminology is treated in [Bognár 1974, chs. III–VI]. The $|1|$ -graded parabolic interpretation is standard [Čap and Slovák 2009, chs. 2–3]. Draper and Meulewaeter [2022, §§2–4] classify inner ideals of finite-dimensional real simple Lie algebras up to automorphism; the abelian nilradicals displayed here belong to that established landscape, and no new classification is claimed. Khare [2017] supplies broader context on standard parabolic subsets, weight-set faces, and associated stabilizers; the local raising-operator argument used here is proved directly. Richardson and Slodowy [1990] is contextual literature on real reductive orbits and minimum vectors, not a source for the compact-image theorem. Quaternionic symplectic notation and dimensions are consistent with [Sedano-Mendoza 2019].

Accordingly, this paper does not claim a new theorem of general Lie theory. Its narrow contribution is:

1. a single, assumption-explicit obstruction package separating continuous neutrality, discrete operator parity, and extremal weight;
2. a complete block-level verification for the fixed $\mathrm{Sp}(32, 32)$ grading;
3. correction of three tempting overstatements: one real eigenvalue is not full hyperbolicity, compact subgroups need not share one fixed Cartan frame, and noncommuting does not imply purely odd;
4. a falsifier and escape taxonomy that makes competing constructions state exactly which hypothesis they leave.

The practical use is diagnostic. It prevents compactness, positivity, and grading preservation from being bundled together by terminology when their stabilizer consequences are incompatible.

12. Conclusion

For a faithful finite-dimensional action, preservation of a positive-definite inner product is equivalent to compact closure of the represented subgroup. This makes three stabilizer obstructions immediate once their hypotheses are typed correctly: A_Z -invariant vectors and Z -commuting operators retain an unbounded grading flow. When $\mathrm{ad}(Z)$ and $dR(Z)$ are real diagonalizable, a maximal (respectively minimal) weight vector retains a positive (respectively negative) unipotent one-parameter subgroup whenever the corresponding eigenspace sum contains X_+ (respectively X_-) with nonzero nilpotent $d\nu$ -image.

In $\mathrm{Sp}(32, 32)$ the required noncompact directions are not inferred from general terminology. They are visible in exact quaternionic blocks: the grading flow has eigenvalues $e^{\pm t}$, and its positive and negative $\mathrm{ad}(Z)$ eigenspaces are 2080-dimensional abelian spaces of square-zero matrices. A compact-reducing involution therefore cannot be grading-even in this module.

The result is a scoped compact-image theorem. Its value lies in making three often conflated obstruction mechanisms, their exact hypotheses, and their honest exits explicit.

Use of generative AI tools

Generative AI tools were used substantially in drafting, literature discovery, formalization and computational support, and adversarial review. The author reviewed the resulting claims and sources and takes full responsibility for the content. The Lean, SageMath, and property-based checks provide independently executable tests of their disclosed narrow targets; they do not transfer authorship or responsibility to the tools and do not substitute for the written proofs.

References

- Bognár, János. *Indefinite Inner Product Spaces*. Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 78. Springer, 1974. See chs. III–VI, pp. 58–146. doi:10.1007/978-3-642-65567-8.
- Čap, Andreas, and Jan Slovák. *Parabolic Geometries I: Background and General Theory*. Mathematical Surveys and Monographs 154. American Mathematical Society, 2009. See chs. 2–3. doi:10.1090/surv/154.
- Draper, Cristina, and Jeroen Meulewaeter. “Inner Ideals of Real Simple Lie Algebras.” *Bulletin of the Malaysian Mathematical Sciences Society* 45 (2022): 2313–2345. See §§2–4. doi:10.1007/s40840-022-01321-0.
- Helgason, Sigurdur. *Differential Geometry, Lie Groups, and Symmetric Spaces*. Graduate Studies in Mathematics 34, corrected reprint. American Mathematical Society, 2001. See chs. III, VI, and IX. doi:10.1090/gsm/034.
- Khare, Apoorva. “Standard Parabolic Subsets of Highest Weight Modules.” *Transactions of the American Mathematical Society* 369 (2017): 2363–2394. doi:10.1090/tran/6710; arXiv:1409.4133.
- Knapp, Anthony W. *Lie Groups: Beyond an Introduction*, 2nd ed. Progress in Mathematics 140. Birkhäuser, 2002. See chs. IV and VI. ISBN 978-0-8176-4259-4.
- Richardson, R. W., and P. J. Slodowy. “Minimum Vectors for Real Reductive Algebraic Groups.” *Journal of the London Mathematical Society* 42 (1990): 409–429. doi:10.1112/jlms/s2-42.3.409.
- Sedano-Mendoza, Manuel. “Isometric Actions of Quaternionic Symplectic Groups.” *Journal of Lie Theory* 29, no. 3 (2019): 755–786. doi:10.5802/jolt.1077.

Paper-specific evidence map

Only the following files are evidence for this paper:

Manuscript and receipts: `good-stable-compactification-no-go.md`; `REPRODUCE.md`; `VERIFICATION.md`.

Exact computational evidence: `evidence/compact_image_obstructions_exact.py`; `evidence/compact_image_obstructions_properties.py`; `evidence/check_lean_receipt.sh`; `evidence/pyproject.toml`; `evidence/uv.lock`; `evidence/checksums.sha256`.

Lean source: `Lean/GUFormalization/CompactImageObstructions.lean`; `Lean/GUFormalization/CompactImageObstructionsAxioms.lean`.

The wider repository contains exploratory and superseded material that is not relied upon here.